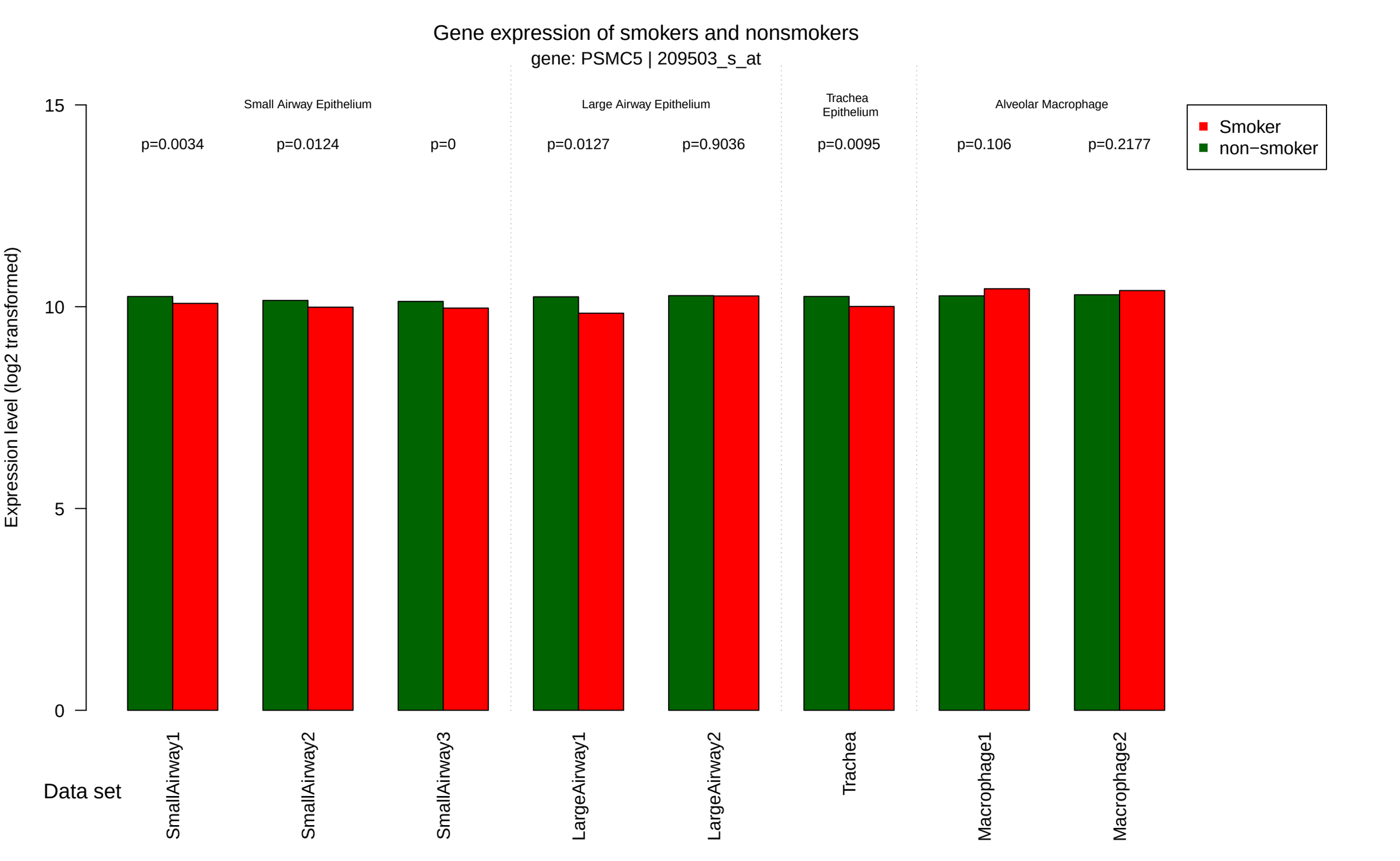
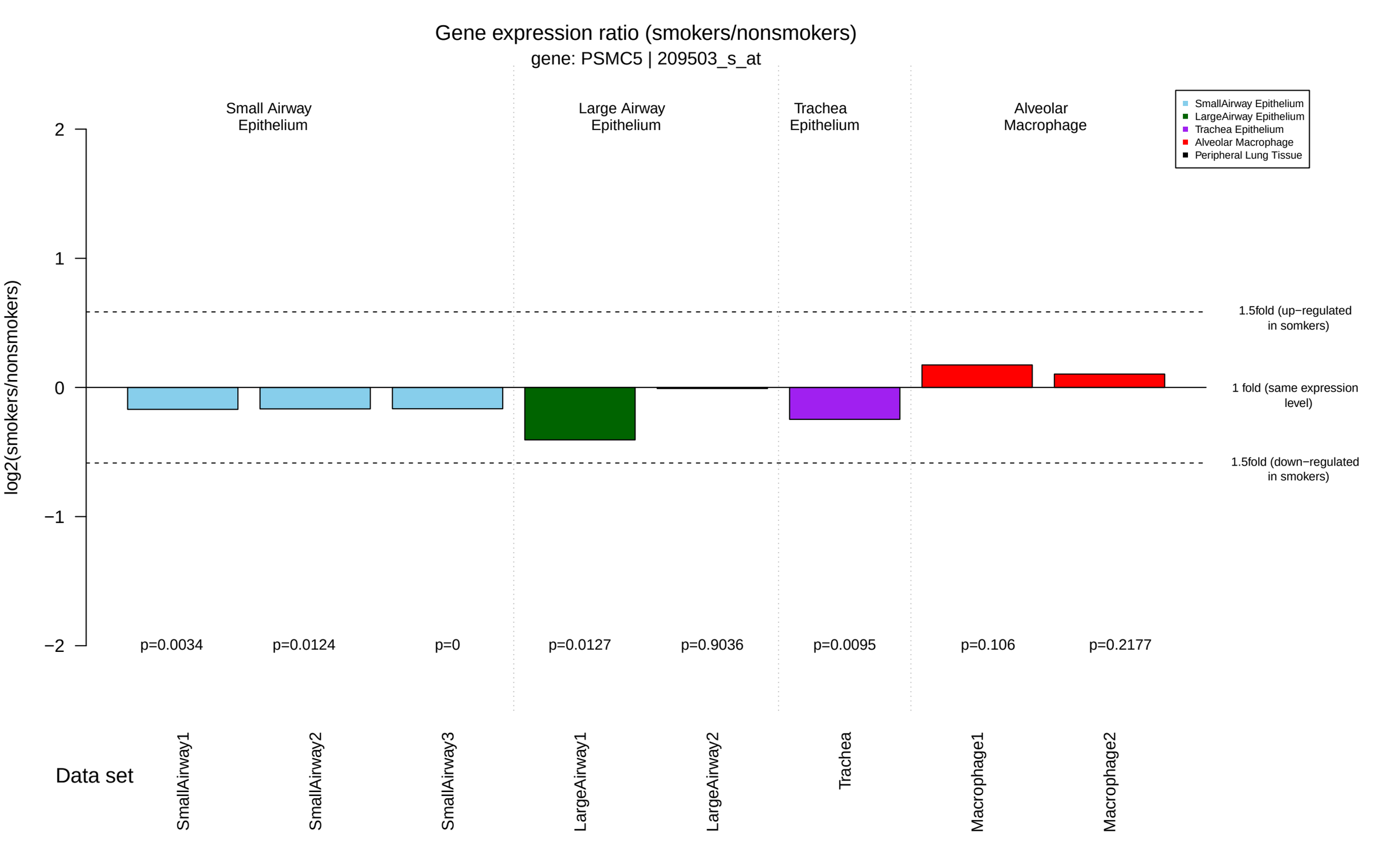
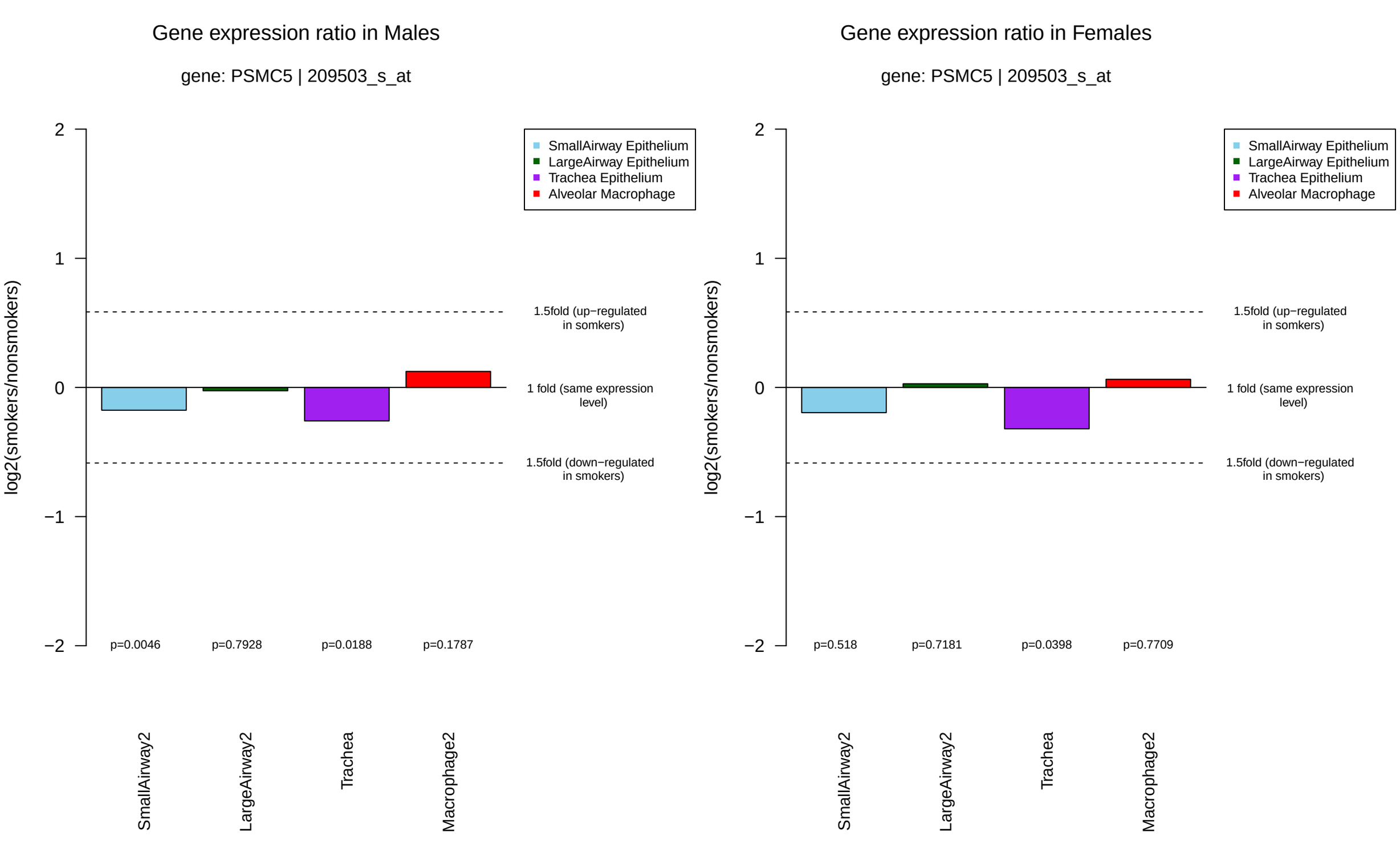
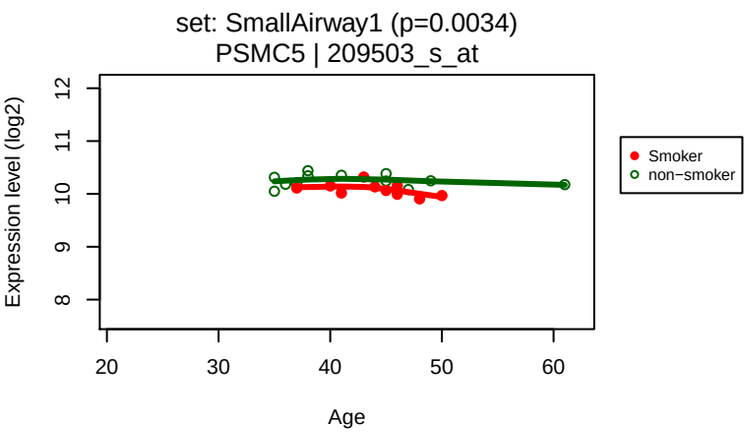
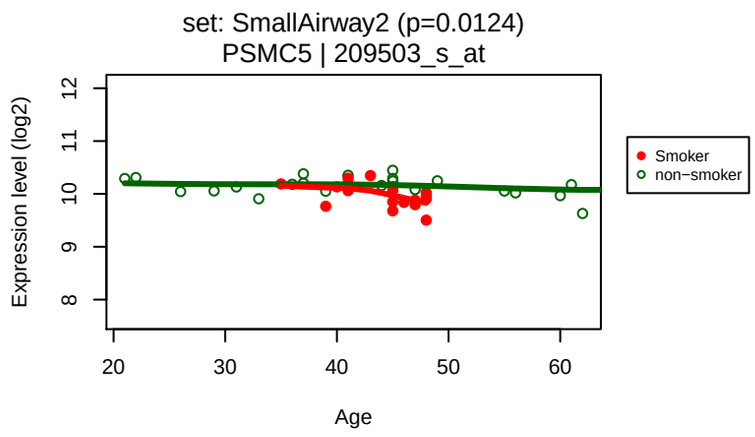
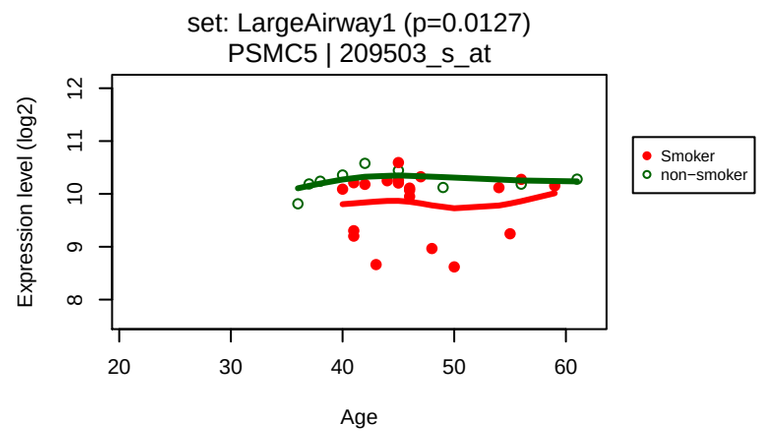

Gene expression of smokers and nonsmokers
gene: PSMC5 | 209503_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0034
p=0.0124
p=0
p=0.0127
p=0.9036
p=0.0095
p=0.106
p=0.2177
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

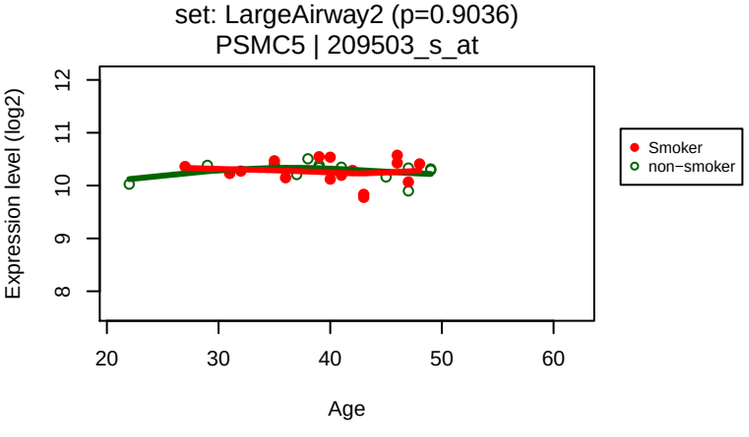

Gene expression ratio (smokers/nonsmokers)
gene: PSMC5 | 209503_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0034
p=0.0124
p=0
p=0.0127
p=0.9036
p=0.0095
p=0.106
p=0.2177
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

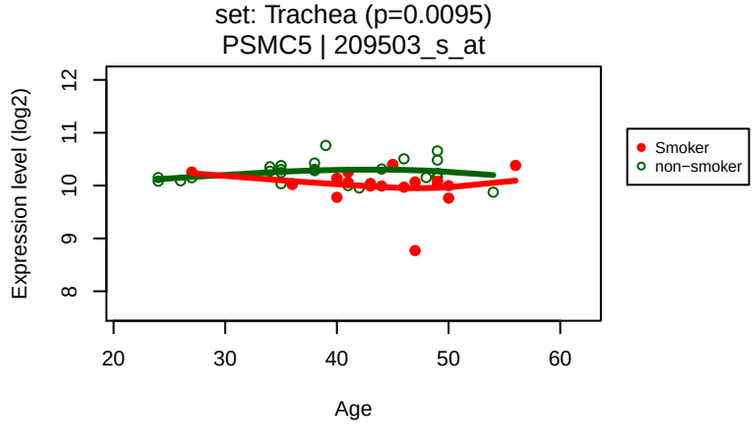

Gene expression ratio in Males
gene: PSMC5 | 209503_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0046
p=0.7928
p=0.0188
p=0.1787
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMC5 | 209503_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.518
p=0.7181
p=0.0398
p=0.7709
SmallAirway2
LargeAirway2
Trachea
Macrophage2

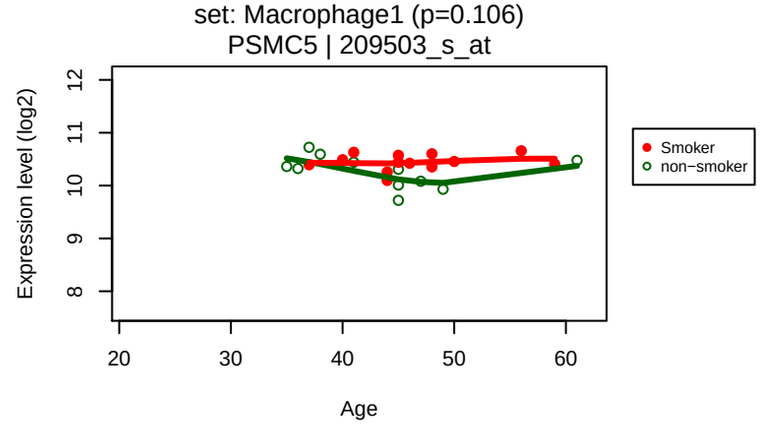

set: SmallAirway1 (p=0.0034)
PSMC5 | 209503_s_at
Expression level (log2)
Age
Smoker
non−smoker

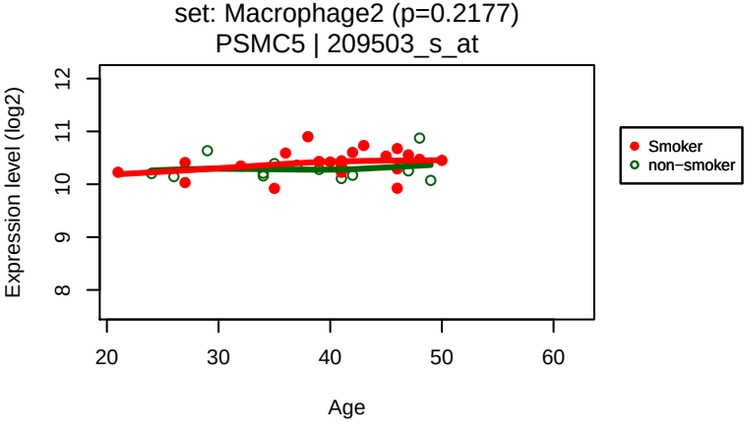

set: SmallAirway2 (p=0.0124)
PSMC5 | 209503_s_at
Expression level (log2)
Age
Smoker
non−smoker

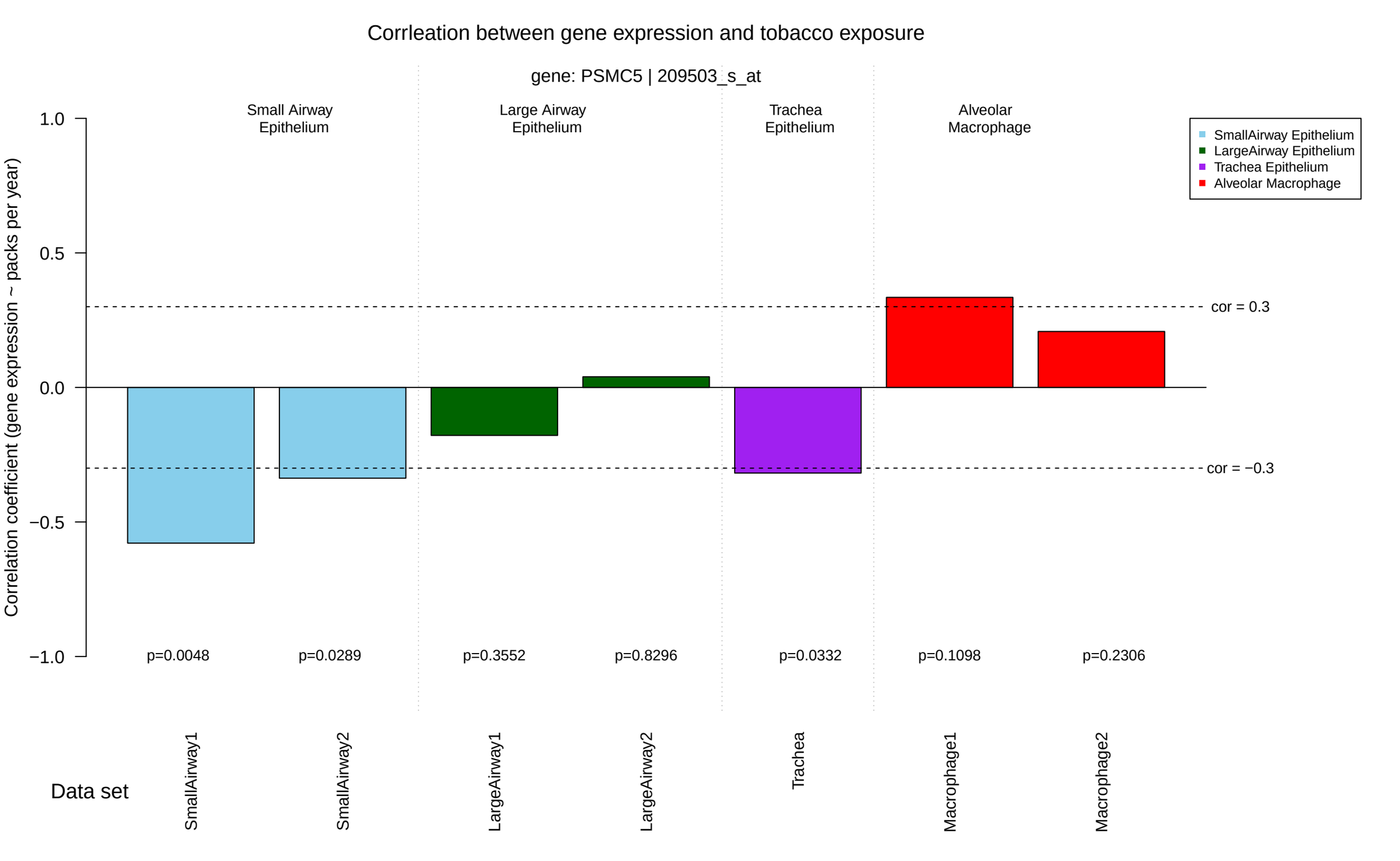

set: LargeAirway1 (p=0.0127)
PSMC5 | 209503_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9036)
PSMC5 | 209503_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0095)
PSMC5 | 209503_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.106)
PSMC5 | 209503_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2177)
PSMC5 | 209503_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMC5 | 209503_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0048
p=0.0289
p=0.3552
p=0.8296
p=0.0332
p=0.1098
p=0.2306
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2